

Econ 677 – paper presentation

No Pass No Drive: Education and Allocation of Time

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Research question:

- Do negative incentives in education improve student outcomes?

What is “No Pass No Drive”(NPND)?

- Set restriction for teenagers to have access to a driver's license.
 - Students must earn their driving privileges by school attendance, academic performance, or behavior.
 - Intention: To motivate academically marginal students, who enjoy the freedom associated with driving, to stay in school or to work harder.
- Cost effective
 - Low Administrative Cost (online notification to traffic authority)
 - Impose no cost to either the state government or the taxpayers
- 1988 West Virginia 2008: 26 states Regulated Age: 15 – 18

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Purpose

- Study the effect of a negative incentive policy on long run educational outcomes.
 - Intuition: NPND should have the biggest effect on individuals who have a preference for driving and are at the margin of dropping out of school.
 - Data: U.S. American Community Survey (ACS)
 - Study the channel through which NPND policy has an effect on education
 - Allocation of time: educational investment, leisure, and work
 - Data: Monitoring the Future (MTF) survey
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Why it is important?

- “Silent Epidemic” phenomenon
 - More than one-third of all public high school students fail to graduate with their class
 - Most studies focus on the effect of performance-based incentives
 - A more direct and cost-effective way to improve student outcomes compared to input-oriented initiatives
 - Too much freedom and too many distraction (Bridgeland et al, 2006)
 - Not very cost effective (Gneezy, Meier and Rey-Biel, 2011)
 - If NPND leads to a decrease in truancy, increase time allocated to studying without penalizing work activities, then it will produce better outcomes than financial incentive programs.
 - Ethical issues involved in conducting randomized controlled trials that would penalize one group of students.
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Do they contribute to the literature?

- Only two papers about the negative incentives on education
 - Lindo, Sanders, and Oreopoulos (2009)
 - ✓ Academic probation → increase dropout rates and GPA
 - Vidal-Fernández (2011)
 - ✓ State interscholastic association rules (student athletes)
 - ✓ Increase in minimum academic standard → 2% increase for graduation probability
 - Contribution
 - Study the effect of negative incentives targeting an activity that students have a preference for, such as driving.
 - The effect of NPND laws on education is arguably homogeneous within subgroups. → It is not so obvious why NPND laws causes increase in dropout rates.
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Main Critique

- Are the effects of NPND laws homogeneous within subgroups?
 - NPND do not affect the utility of staying in school.
 - They make the outside option of dropping out less attractive if students have a preference for driving.
 - Counter Argument
 - ✓ Some students may no longer be able to attend school if they lose their driving privileges.
 - ✓ Some students may drop out of school if the law imposes financial burdens (due to a decrease in hours spend working).
 - Critique
 - ✓ Can appeal for an exemption based on personal reasons
 - Not every state allows this kind of exemption.
 - Need to check validity based on the exemption
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Model for the effect of NPND laws on education outcomes

➤ Outcome variables: Graduation rates & Years of schooling

$$E_{isc} = \beta_1 Treatment_{sc} + \beta_2 X_{isc} + \beta_3 R_{sc} + S + B + \epsilon_{isc}$$

$Treatment_{sc}$: indicating whether the individual belongs to the treated cohort c in state of birth s

= 1 for individuals younger than 13 in the year the law was passed in a state

= 0 for individuals older than 18 in the year the law was passed in a state

X_{isc} : controls for gender, race, and Standard Metropolitan Statistical Area (SMSA)

R_{sc} : state-specific demographic, economic, and education controls

S, B : state of birth and year of birth fixed effects

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Main Critique

- Standard errors are clustered on both state and year (Bertrand et al., 2004)
- Problem: understate the correct standard error
 - Study 92 paper that use DD
 - 80 paper with potential clustering problem
 - Found a significant effect up to 45% of the time (at 5% level)
- **Should cluster only on state**
 - **Allow errors to be correlated within states across time**

Minor Critique

- Use state of birth as identifier to avoid induced migration
- **Compositional change problem**
- **Should restrict the sample to id residing in same state of birth**

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Identifying assumption:

- ❑ Education outcomes do not vary systematically across cohorts in the treatment and control states over time.
 - Potential internal validity threats and Strategy to check for them
 - 1) Educational outcomes were reacting to other laws that were being implemented around the same time.
 - Focus on minimum school entrance age laws and compulsory attendance laws
 - 2) Mean reversion if there was a downward trend in treatment states
 - Include state-specific linear time trends
 - 3) Policy endogeneity (increase in teen car accident fatality rates)
 - Controlling for two additional state level traffic control variables (vehicle miles traveled and total motor vehicle fatalities among 15-17 years old)
 - Placebo regressions among older cohorts not affected by NPND
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Model for how NPND affects time allocation

- Outcome variables (time allocation): working hours, leisure, homework, and driving
 - $Y_{ist} = \alpha_0 + \alpha_1 NPND_{st} + \alpha_2 X_{ist} + \alpha_3 Z_{st} + S + T + \epsilon_{ist}$
 $NPND_{st} = 1$ if the state s has the NPND law in place at time t
 X_{ist} : Individual characteristic
(age, max. parental education, race, gender)
 Z_{st} : Potentially relevant time varying state-specific controls
(macroeconomic, education, and traffic related variables)
 S, T : State and time fixed effects
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Identifying assumption:

- ❑ There are no unobserved changes in variables related to both student outcomes and NPND laws that are differentially affecting treatment and control states.
- Potential internal validity threats and Strategy to check them
- 1) If NPND laws were enacted together with other traffic laws, we would find a decrease in leisure and in probability of driving for teenagers which may not entirely be attributed to NPND laws.

Robustness checks:

- Include state-specific linear time trends to capture unobserved characteristics at the state level
- Results are robust to Graduate Driver Licensing and Seatbelt use laws
→ results are not presented
- Introduce education controls that affect education
 - expenditure/pupil, teacher salary, pupil/teacher ratio

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Main Critique

- The same as the first model: clustering problem
- MTF only records information for non-dropouts.
 - Implication: Selection bias
 - Understates the true coefficient of time allocation to school
 - Not a big problem: conservative estimates

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Empirical results: NPND law on HS graduation and educ. attainment

Outcome		Only Treatment	With Fixed Effect	All	Males Only	Females Only
Graduation	Base	-0.030***	0.011***	0.009***	0.010***	0.008***
		(0.001)	(0.002)	(0.002)	(0.003)	(0.002)
	With Time Trends			0.010***	0.015***	0.005
				(0.004)	(0.006)	(0.004)
Attainment	Base	-0.127***	0.001	0.036**	0.061***	0.011
		(0.008)	(0.021)	(0.015)	(0.021)	(0.019)
	With Time Trends			0.054**	0.091***	0.019
				(0.025)	(0.033)	(0.032)

***p<0.01, ** p<0.05, * p<0.1

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Empirical results: effect of NPND laws on education – robustness check

	Attainment				Graduation			
	Baseline	MESL	CAL	Placebo	Baseline	MESL	CAL	Placebo
Males Only								
Treatment	0.091***	0.079**	0.094***	0.003	0.015***	0.018***	0.014**	-0.002
	(0.033)	(0.035)	(0.033)	(0.032)	(0.006)	(0.006)	(0.006)	(0.005)
MESL/CAL		-0.013	-0.01			0.001	0.005***	
		(0.009)	(0.012)			(0.001)	(0.002)	
Black Males Only								
Treatment	0.344***	0.414***	0.349***	-0.127	0.051***	0.074***	0.051***	-0.009
	(0.096)	(0.105)	(0.096)	(0.102)	(0.018)	(0.020)	(0.018)	(0.016)
MESL/CAL		0.017	-0.012			0.005	0.006	
		(0.024)	(0.032)			(0.005)	(0.005)	

***p<0.01, ** p<0.05, * p<0.10

MESL: Minimum entry age law

CAL: Compulsory attendance laws

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Empirical results: NPND law on education, employment and leisure

	All	Male	Female	Black	Non-Blacks
Grades	0.024 (0.047)	0.077 (0.060)	-0.022 (0.053)	0.028 (0.093)	0.011 (0.054)
Prob. of skipping school	-0.018** (0.009)	-0.015 (0.011)	-0.020** (0.009)	-0.020 (0.017)	-0.016* (0.010)
Hours spent doing homework a day	0.197 (0.158)	0.359** (0.161)	0.062 (0.215)	0.964*** (0.223)	-0.070 (0.181)
Hours worked a day	-0.105* (0.054)	-0.145** (0.063)	-0.077 (0.065)	-0.180** (0.078)	-0.074 (0.059)
Going out on a date	-0.067*** (0.021)	-0.052* (0.027)	-0.079*** (0.023)	-0.027 (0.025)	-0.081*** (0.025)
Going out for parties	-0.023 (0.023)	0.031 (0.026)	-0.063** (0.026)	-0.015 (0.031)	-0.025 (0.023)
Watching TV	-0.030 (0.033)	-0.017 (0.039)	-0.043 (0.037)	-0.194*** (0.063)	-0.022 (0.034)

***p<0.01, ** p<0.05, * p<0.1

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Empirical results: summary

- NPND laws have a positive effect on education outcomes among boys and blacks, but not girls.
- It led to a 2.7 percentage point increase in average educational attainment among black males and a 6.4 percentage point increase in the probability of graduating from high school.
- This result is particularly relevant because dropout rates are strikingly high among blacks and males.
- As for channel through which NPND policy has an effect on education:
 - NPND laws are effective in reducing truancy and increased time allocated to homework, mainly among blacks, at the expense of leisure and employment activities.

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Conclusion

- Overall, good paper
 - Contribute to the literature (negative incentives)
 - Research question (actual problem: “Silent Epidemic” phenomenon)
 - Identifying assumption and robustness check: “well conducted”
- Main critiques:
 - Clustering choice: leads to over-rejection
 - Sample selection (MTF data)
 - Robustness check for Identifying assumption 2.
 - ✓ Effect of other traffic laws on time allocation

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THANK YOU!!!

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